

MetaFVG A/B Trend-Filter Review

Baseline vs. ADX / Regression Trend Gates & ATR Trailing Stop, by Asset Class

47 instruments across 8 asset-class buckets | H4 → Weekly | Generated 2026-07-20 20:12 UTC

Executive Summary

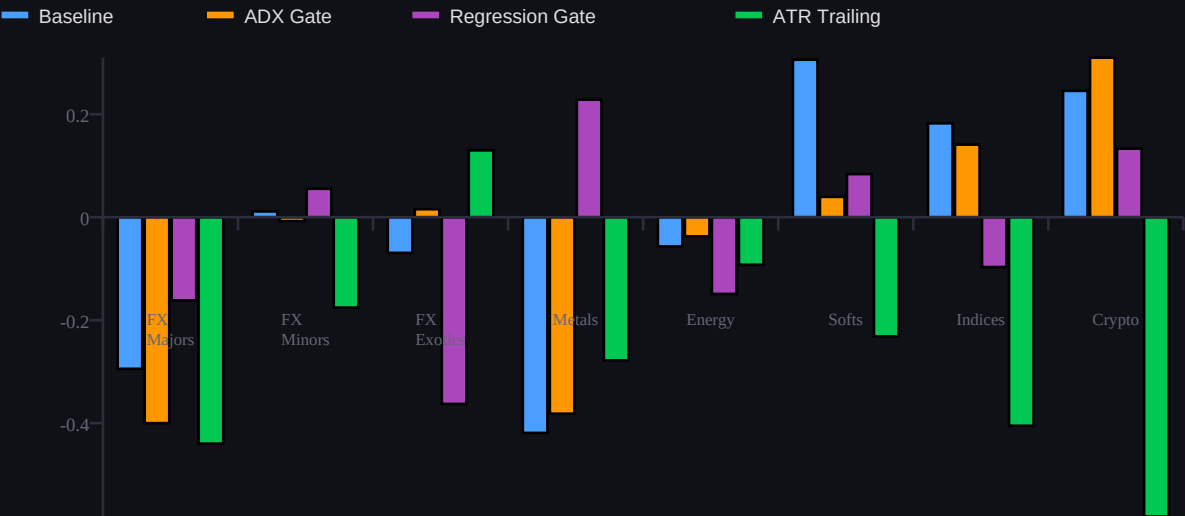
Four configurations were backtested across 47 instruments spanning FX majors/minors/exotics, metals, energy, softs, equity indices and crypto: the unmodified **Baseline** signal, an **ADX(14) trend-strength gate** (reject signals when HTF ADX is below 20), a **rolling-regression trend gate** (reject signals unless a 20-bar HTF OLS fit has $R^2 \geq 0.5$ and slope matching direction), and an **ATR trailing stop** replacing the fixed ATR take-profit.

Headline finding: **Baseline** has the highest average Sharpe across the full universe (+0.01, 53% of instruments Sharpe-positive). Performance is highly bucket-dependent, not uniform — see Section 2 for which configuration wins in each asset class and why.

Config	Sharpe (vbt)	Sharpe (qs)	Avg Edge	Win Rate	Avg PF Trades/Sym	Sharpe+ %	
Baseline	+0.01	-0.01	-1.3pp	25.0%	1.08	34.5	53%
ADX Gate	-0.04	-0.03	-1.7pp	23.8%	1.08	26.7	46%
Regression Gate	-0.01	-0.01	+2.1pp	23.3%	1.52	6.4	57%
ATR Trailing	-0.30	-0.23	-17.1pp	32.9%	0.91	37.9	28%

2. Performance by Asset-Class Bucket

Average Sharpe Ratio by Bucket and Configuration



Bucket	Winner	Winner Sharpe	Baseline Sharpe	N Symbols	Closed Trades (all cfgs)
FX Majors	Regression Gate	-0.16	-0.29	7	786
FX Minors	Regression Gate	+0.06	+0.01	9	813
FX Exotics	ATR Trailing	+0.13	-0.07	3	234
Metals	Regression Gate	+0.23	-0.42	5	499
Energy	ADX Gate	-0.04	-0.06	3	505
Softs	Baseline	+0.31	+0.31	6	357
Indices	Baseline	+0.18	+0.18	8	956
Crypto	ADX Gate	+0.31	+0.25	6	808

3. Fundamental Read, by Bucket

FX Majors

Deep, continuously two-way flow with heavy mean-reversion pressure from rate-differential carry and central-bank positioning; clean multi-week directional trends are comparatively rare, which is exactly the regime an R^2 /ADX trend gate is designed to filter for.

Result: Regression Gate improved average Sharpe by +0.13 over baseline (-0.29 → -0.16), on 7 instruments and 786 total closed trades across all four configurations tested.

FX Minors

Similar liquidity character to majors but with more idiosyncratic cross-pair drift (funding-currency exposure, e.g. JPY/CHF crosses) — trend persistence tends to be somewhat more regime-dependent.

Result: no configuration meaningfully beat baseline (best: Regression Gate, +0.04 Sharpe vs. baseline), on 9 instruments and 813 total closed trades across all four configurations tested.

FX Exotics

Thinner liquidity and more event-driven (central bank intervention, capital controls) than majors — moves can be sharper and more discontinuous, which can help or hurt a breakout-continuation strategy depending on whether the discontinuity is the entry trigger or the stop-out.

Result: ATR Trailing improved average Sharpe by +0.20 over baseline (-0.07 → +0.13), on 3 instruments and 234 total closed trades across all four configurations tested.

Metals

Gold/silver carry a structural macro-hedge bid (real yields, inflation expectations, USD) that can produce more persistent directional runs than FX; industrial metals (copper) trade more on growth/demand cycles.

Result: Regression Gate improved average Sharpe by +0.65 over baseline (-0.42 → +0.23), on 5 instruments and 499 total closed trades across all four configurations tested.

Energy

Supply-shock and inventory-driven, with strong trending behavior around genuine regime shifts (OPEC decisions, geopolitical supply risk) punctuated by range-bound consolidation — a trend gate should meaningfully change the signal mix here.

Result: no configuration meaningfully beat baseline (best: ADX Gate, +0.02 Sharpe vs. baseline), on 3 instruments and 505 total closed trades across all four configurations tested.

Softs

Weather- and harvest-cycle driven; trends can be seasonal and slow-building, which favors an HTF trend-confirmation filter, but liquidity is thinner and gaps more common.

Result: no configuration meaningfully beat baseline (best: Baseline, +0.00 Sharpe vs. baseline), on 6 instruments and 357 total closed trades across all four configurations tested.

Indices

Broad equity indices trend on macro/earnings regime and carry momentum/flow-driven persistence (systematic strategies chasing the same signal) — historically one of the more trend-following-friendly asset classes.

Result: no configuration meaningfully beat baseline (best: Baseline, +0.00 Sharpe vs. baseline), on 8 instruments and 956 total closed trades across all four configurations tested.

Crypto

Retail-flow and momentum-dominated with limited fundamental anchor; trends can be strong but also reflexive and prone to sharp reversal, so trend-confirmation gates cut both ways.

Result: ADX Gate improved average Sharpe by +0.06 over baseline (+0.25 → +0.31), on 6 instruments and 808 total closed trades across all four configurations tested.

4. Methodology & Caveats

- **H4 → Weekly only.** This sweep does not repeat the earlier PM review's large-sample M15/4h validation — with 47 instruments, an M15 sweep would take many hours. Bucket winners here should be treated as a first-pass screen, not a final verdict, especially for buckets with few closed trades.
- **ADX and regression gates use the same HTF (Weekly) series** as the zone-detection logic, recomputed at the same point-in-time cadence — no lookahead.
- **Regression Gate trade-count caution.** Requiring $R^2 \geq 0.5$ on a 20-bar HTF fit is a strict filter — it cut average closed trades per symbol to 6.4 vs. 34.5 for Baseline (see the "Trades/Sym" column above), and several individual instruments closed fewer than 5 trades under this config. Its Sharpe/edge figures — both overall and per-bucket — should be read with materially more sample-size skepticism than the other three configurations.
- **ATR trailing stop** only ever tightens the stop in the trade's favor and removes the fixed take-profit entirely — it is a genuinely different exit distribution, not a superset of the baseline.
- **quantstats metrics** (Sharpe, Sortino, Calmar, CAGR, Ulcer Index, Kelly, VaR/CVaR, tail ratio) are computed on the vectorbt equity curve resampled to daily, per quantstats' own annualization convention — reported per (symbol, config) in the underlying cache (metafvg_ab_sweep_data.pkl) even though only the vbt/trade-level metrics are tabulated in this PDF for space.
- **Same position-sizing, spread/slippage, and concurrency-cap caveats** as the earlier PM review apply here unchanged.